# T-SF32-Display_V1.0 Power consumption test

Date: 2026_06_26
Test Board: T-SF32-Display_V1.0 && T-SF32-Keyboard_V1.0
Software: menu_app_v1.0
Test organization: LILYGO

## 1. Test Environment Description

1.1 Test environment(测试环境)

| Test Tools | Sampling speed | Test Methodology | Input voltage | Test environment temperature |
|---|---|---|---|---|
| BLU939 | 100000 times/s | Battery Interface Measurement | +4.0V | 25°C/77°F |

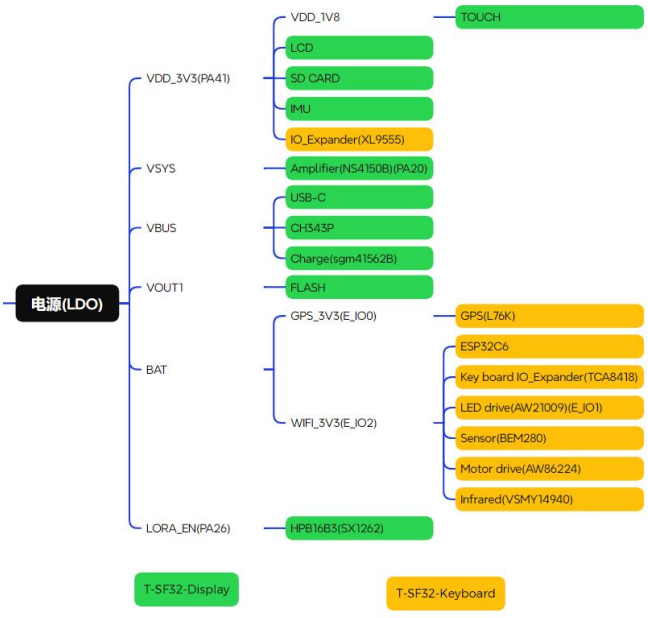
1.2 Test device model(测试器件型号)

1.2.1 T-SF32-Display

| Test deviece | Screen | Lora | BlueTooth | Loudspeaker | IMU | TF |
|---|---|---|---|---|---|---|
| Component model | H0216F002AM | HPB16B3(SX1262) | SIFLI | NS4150B | BHI260AP | MicroSD |

1.2.2 T-SF32-Keyboard

| Test deviece | Slave MCU | Keyboard | Key led | GPS | Sensor | Motor | IR |
|---|---|---|---|---|---|---|---|
| Component model | ESP32C6 | TCA8418 | AW21009QNR | L76K | BME280 | AW86224AFCR | VSMY14940 |




Power Control Diagram

## 2. Test Physical Object Image

## 3. Power Consumption Test

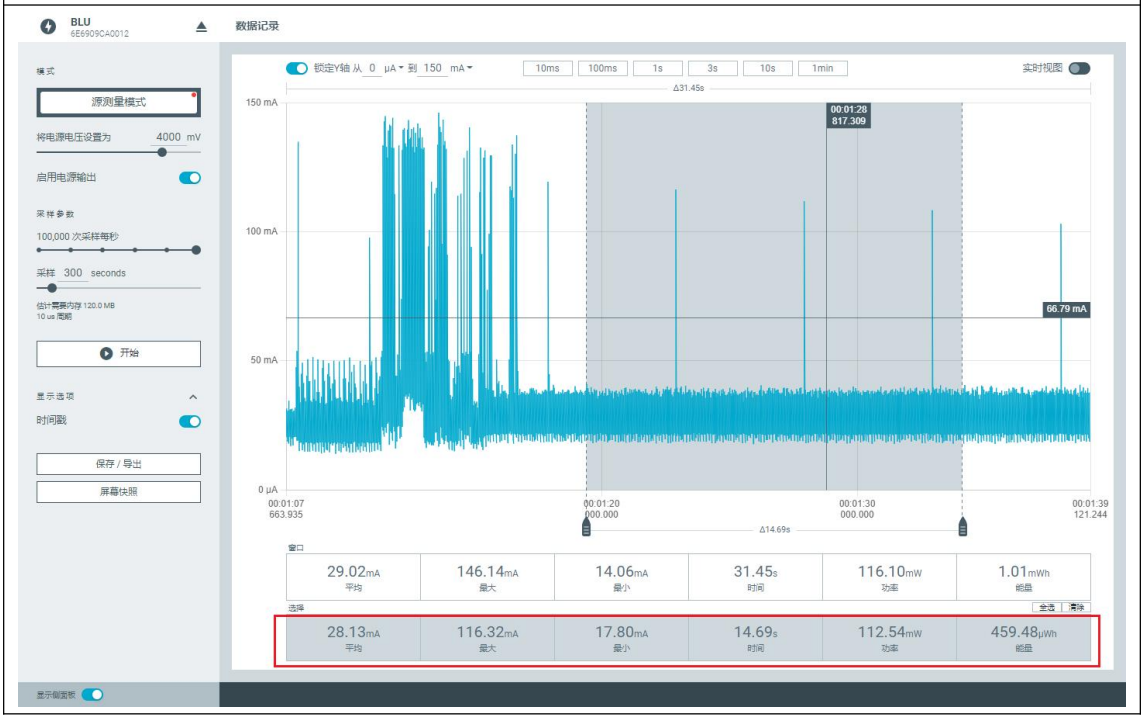
-The screen is set to 100 brightness level when turned on.

-LORA transmission is the maximum transmission power, while LORA reception is in the enhanced reception mode.

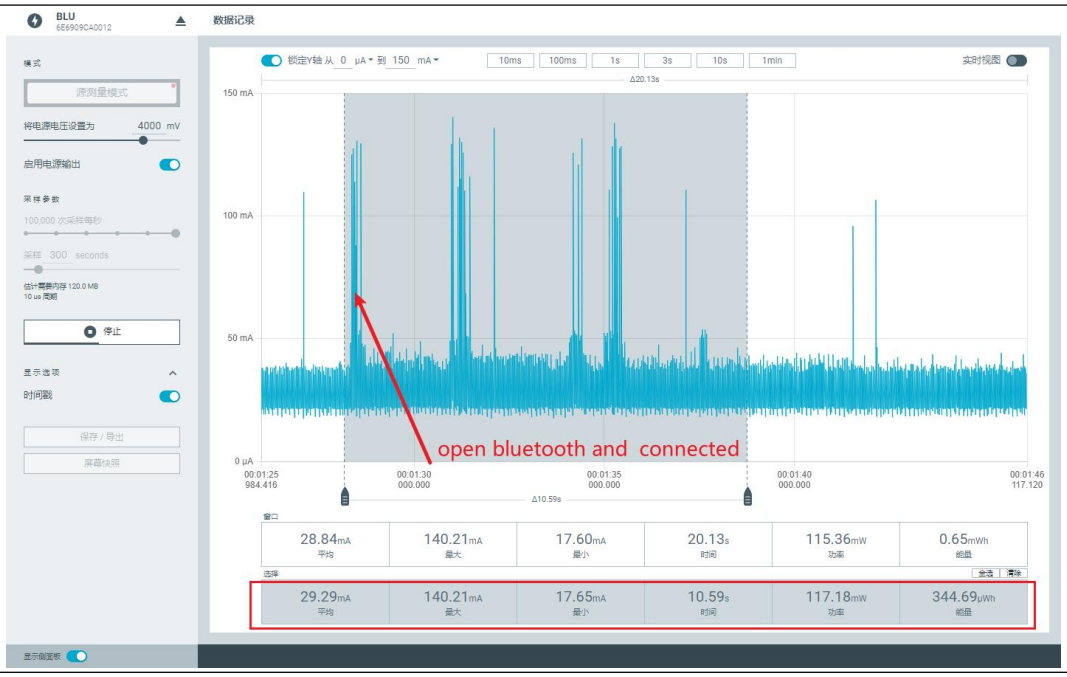
-The audio playback volume is set at 50% level.

-This firmware uses the STA mode for the ESP32C6.

| SFLB52X | LCD | TOUCH | SD CARD | IMU | NS4150B | LORA | BLUETOOTH |
|---|---|---|---|---|---|---|---|
| Active | ON | ON | ON | SLEEP | OFF | SLEEP | OFF |
| WIFI | GPS | KEYBOARD | KEY_LED | SENSOR | MOTOR | INFRARED | |
| OFF | OFF | OFF | OFF | OFF | OFF | OFF | |
| Test result: average 28.13mA | | | | | | | |



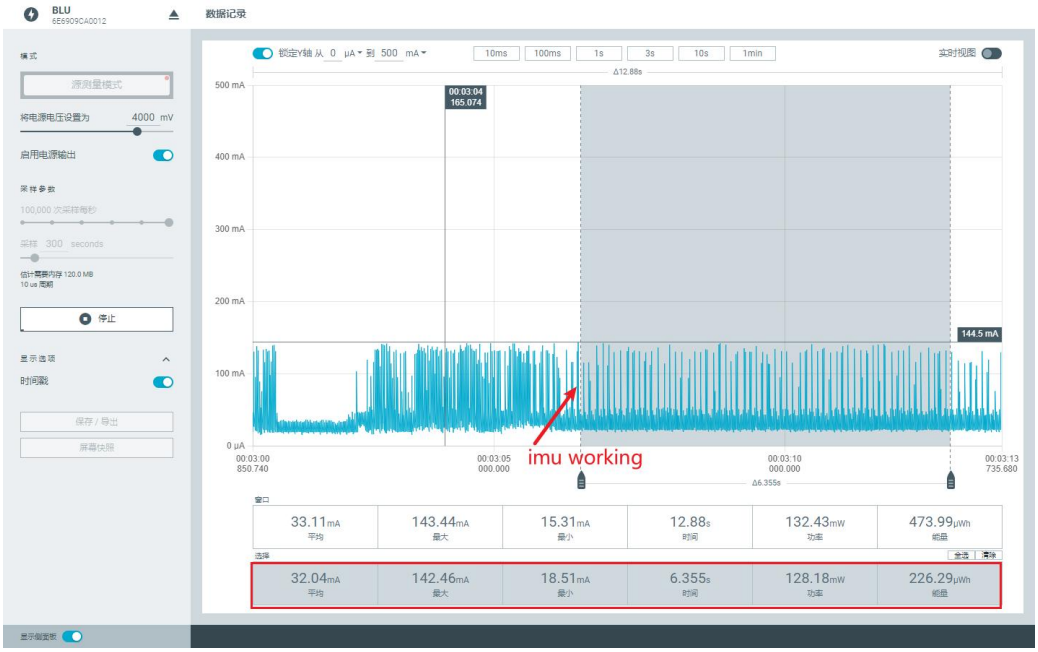
| SFLB52X | LCD | TOUCH | SD CARD | IMU | NS4150B | LORA | BLUETOOTH |
|---|---|---|---|---|---|---|---|
| Active | ON | ON | ON | SLEEP | OFF | SLEEP | ON |
| WIFI | GPS | KEYBOARD | KEY_LED | SENSOR | MOTOR | INFRARED | |
| OFF | OFF | OFF | OFF | OFF | OFF | OFF | |
| Test result: average 29.29mA | | | | | | | |




| SFLB52X | LCD | TOUCH | SD CARD | IMU | NS4150B | LORA | BLUETOOTH |
|---|---|---|---|---|---|---|---|
| Active | ON | ON | ON | SLEEP | ON | SLEEP | ON |
| WIFI | GPS | KEYBOARD | KEY_LED | SENSOR | MOTOR | INFRARED | |
| OFF | OFF | OFF | OFF | OFF | OFF | OFF | |
| Test result: average 45.42mA | | | | | | | |



| SFLB52X | LCD | TOUCH | SD CARD | IMU | NS4150B | LORA | BLUETOOTH |
|---|---|---|---|---|---|---|---|
| Active | ON | ON | ON | ON | OFF | SLEEP | OFF |
| WIFI | GPS | KEYBOARD | KEY_LED | SENSOR | MOTOR | INFRARED | |
| OFF | OFF | OFF | OFF | OFF | OFF | OFF | |
| Test result: average 32.04mA | | | | | | | |




| SFLB52X | LCD | TOUCH | SD CARD | IMU | NS4150B | LORA | BLUETOOTH |
|---|---|---|---|---|---|---|---|
| Active | ON | ON | ON | SLEEP | OFF | TX | OFF |
| WIFI | GPS | KEYBOARD | KEY_LED | SENSOR | MOTOR | INFRARED | |
| OFF | OFF | OFF | OFF | OFF | OFF | OFF | |
| Test result: average 47.62mA | | | | | | | |



| SFLB52X | LCD | TOUCH | SD CARD | IMU | NS4150B | LORA | BLUETOOTH |
|---|---|---|---|---|---|---|---|
| Active | ON | ON | ON | SLEEP | OFF | RX | OFF |
| WIFI | GPS | KEYBOARD | KEY_LED | SENSOR | MOTOR | INFRARED | |
| OFF | OFF | OFF | OFF | OFF | OFF | OFF | |
| Test result: average 60.45mA | | | | | | | |

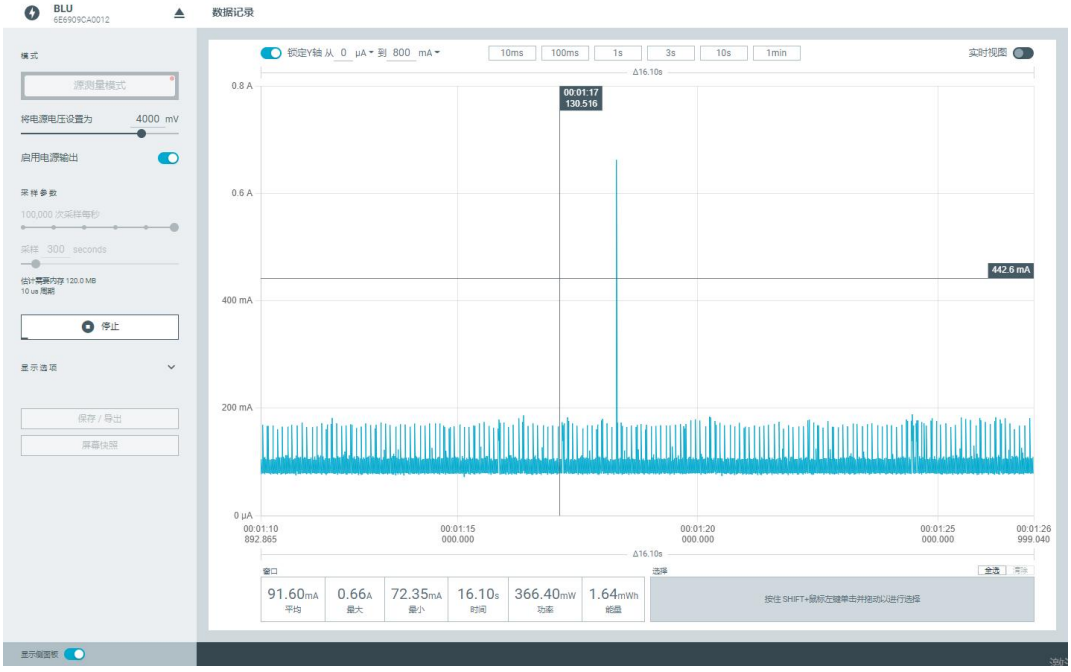



| SFLB52X | LCD | TOUCH | SD CARD | IMU | NS4150B | LORA | BLUETOOTH |
|---|---|---|---|---|---|---|---|
| Active | OFF | ON | ON | SLEEP | OFF | SLEEP | OFF |
| WIFI | GPS | KEYBOARD | KEY_LED | SENSOR | MOTOR | INFRARED | |
| OFF | OFF | OFF | OFF | OFF | OFF | OFF | |
| Test result: average 22.99mA | | | | | | | |



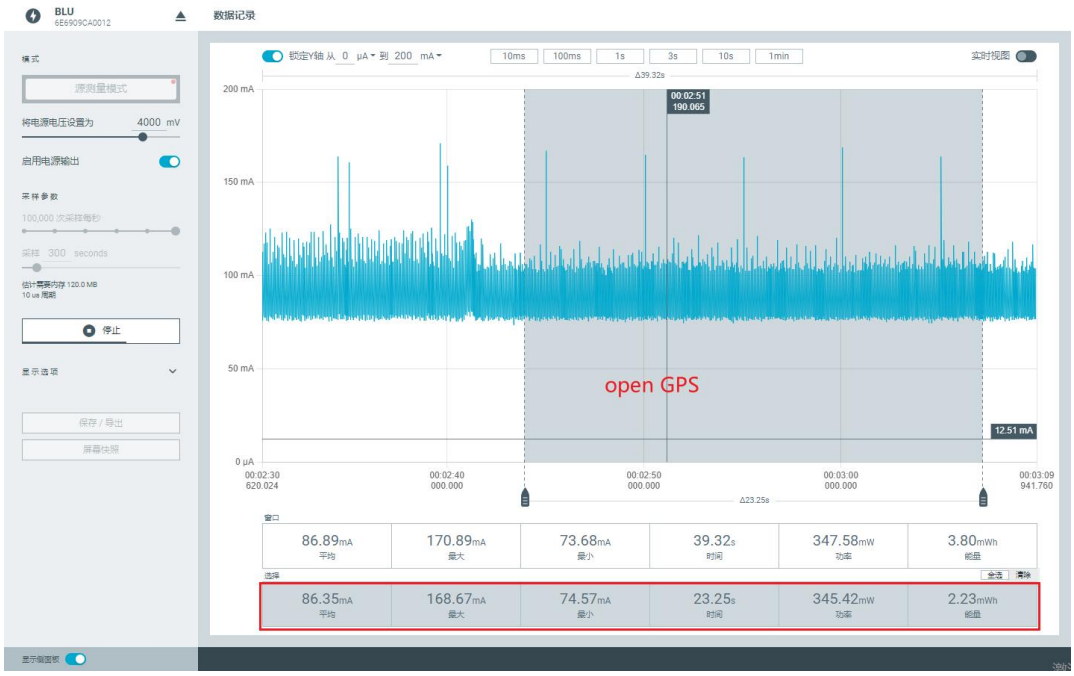

| SFLB52X | LCD | TOUCH | SD CARD | IMU | NS4150B | LORA | BLUETOOTH |
|---|---|---|---|---|---|---|---|
| Active | ON | ON | ON | SLEEP | OFF | SLEEP | OFF |
| WIFI | GPS | KEYBOARD | KEY_LED | SENSOR | MOTOR | INFRARED | |
| ON | OFF | ON | ON | ON | OFF | OFF | |

Test result: average 91.60mA




| SFLB52X | LCD | TOUCH | SD CARD | IMU | NS4150B | LORA | BLUETOOTH |
|---|---|---|---|---|---|---|---|
| Active | ON | ON | ON | SLEEP | OFF | SLEEP | OFF |
| WIFI | GPS | KEYBOARD | KEY_LED | SENSOR | MOTOR | INFRARED | |
| OFF | ON | OFF | OFF | OFF | OFF | OFF | |

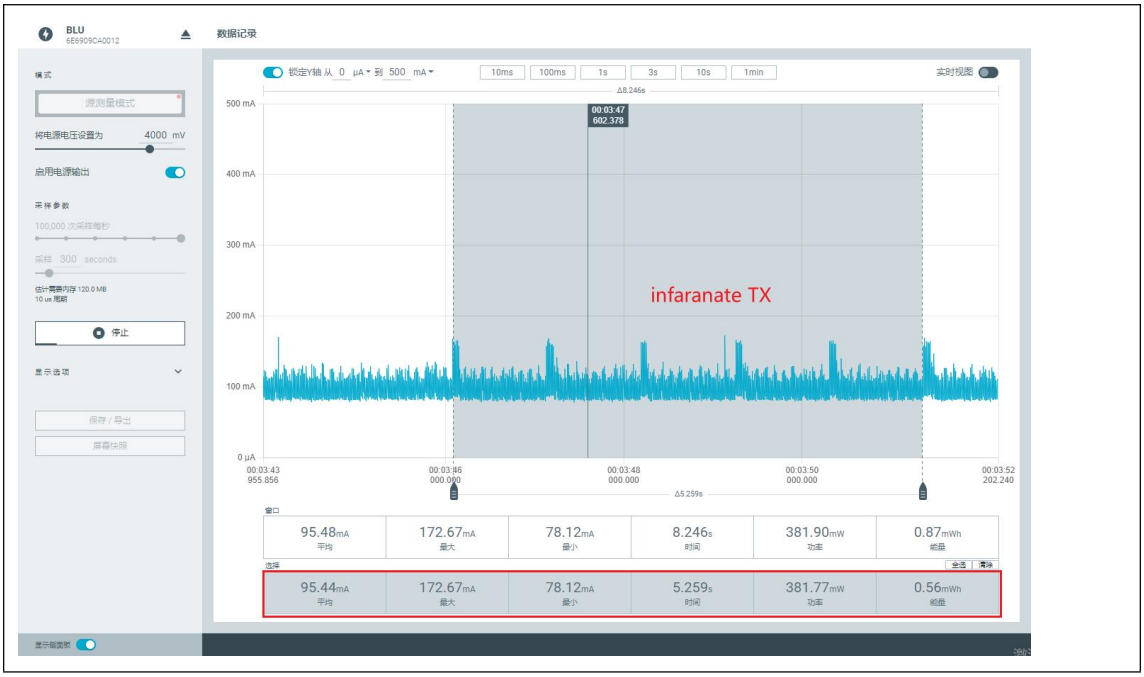
Test result: average 86.35mA




| SFLB52X | LCD | TOUCH | SD CARD | IMU | NS4150B | LORA | BLUETOOTH |
|---|---|---|---|---|---|---|---|
| Active | ON | ON | ON | SLEEP | OFF | SLEEP | OFF |
| WIFI | GPS | KEYBOARD | KEY_LED | SENSOR | MOTOR | INFRARED | |
| OFF | OFF | OFF | OFF | OFF | ON | OFF | |
| Test result: average 193.08mA | | | | | | | |




| SFLB52X | LCD | TOUCH | SD CARD | IMU | NS4150B | LORA | BLUETOOTH |
|---|---|---|---|---|---|---|---|
| HIBERNATE | OFF | OFF | OFF | OFF | OFF | OFF | OFF |
| WIFI | GPS | KEYBOARD | KEY_LED | SENSOR | MOTOR | INFRARED | |
| OFF | OFF | OFF | OFF | OFF | OFF | OFF | |
| Test result: average 95.44mA | | | | | | | |




| SFLB52X | LCD | TOUCH | SD CARD | IMU | NS4150B | LORA | BLUETOOTH |
|---|---|---|---|---|---|---|---|
| HIBERNATE | OFF | OFF | OFF | OFF | OFF | OFF | OFF |
| WIFI | GPS | KEYBOARD | KEY_LED | SENSOR | MOTOR | INFRARED | |
| OFF | OFF | OFF | OFF | OFF | OFF | OFF | |

Test result: average 164uA



| SFLB52X | LCD | TOUCH | SD CARD | IMU | NS4150B | LORA | BLUETOOTH |
|---|---|---|---|---|---|---|---|
| Shotdown | OFF | OFF | OFF | OFF | OFF | OFF | OFF |
| WIFI | GPS | KEYBOARD | KEY_LED | SENSOR | MOTOR | INFRARED | |
| OFF | OFF | OFF | OFF | OFF | OFF | OFF | |
| Test result: average 0.87uA | | | | | | | |

